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Oak Ridge National Laboratory RSI Internship Reflection

During my appointment at Oak Ridge National Laboratory, I learned how to build automated control systems for the Uranium Science and Technology Center under the tutelage of senior instrumentation and controls engineers. The focal point of this experience was the use and development of Siemens Programmable Logic Controllers (PLCs) and Human Machine Interfaces (HMIs) for the automation of controls. Under the guidance of my mentor, I learned how to develop Siemens PLC systems using Ladder Logic style programming, how to develop HMIs with intuitive design, how to create reusable HMI faceplates, and how to interface with simple remote sensors and analog/digital inputs and outputs. I was also exposed to the engineering process which includes design, component selection, programming, testing, and release of PLC software. My primary responsibility was to develop automation for a gas-sampling system. This was done using a Siemens S7-1200 PLC, TIA Portal development software, and a Siemens Simatic HMI (Unified Comfort). This project allowed me to apply my computer science, and specifically embedded systems background to a physical automation system. Moreover, I was able to learn how Industrial Controls are designed, tested, and presented to operators.

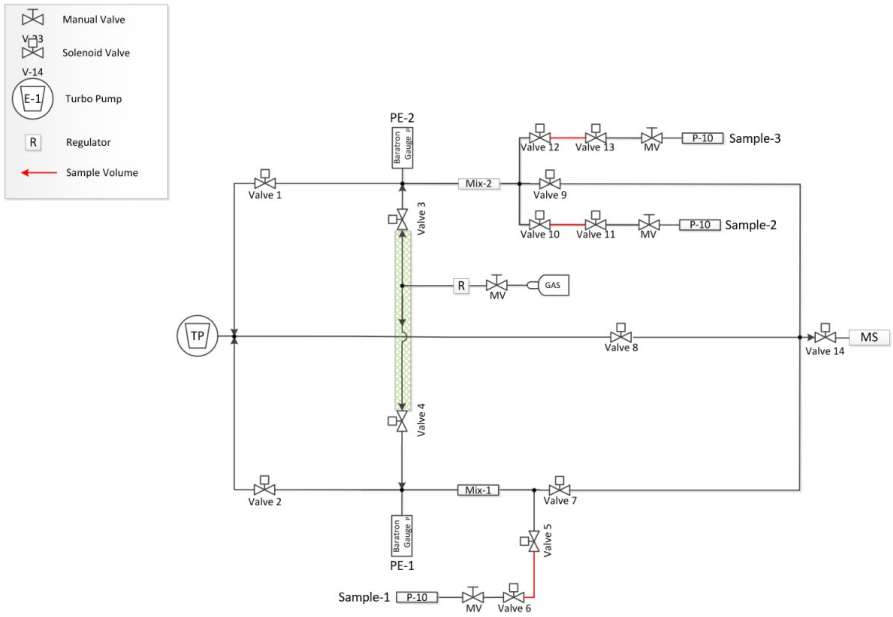


Figure 1. Gas sampling system schematic

System Overview

For my project I was given a schematic diagram (see figure 1) as well as a narrative that details the expected behavior of the system. My job was to translate these resources into a functioning

system. The first step was to recreate the schematic diagram into a screen for the HMI using TIA Portal. This screen serves as the main control panel for the sampling system. After the visual structure for the control panel was established, I then translated the narrative into functional logic using ladder. I developed function blocks for the following pre-sampling modes: Initial vacuum clearance, Sample 1 gas cleanout, sample 2 gas cleanout, sample 3 gas cleanout, close all valves. I then moved onto doing the same thing for the following sampling modes: Sample 1 mix, sample 2 mix, sample 3 mix, sample to mass spectrometer. How these modes work together to perform a complete sampling operation is as follows.

- Put all valves in a known state – Close All
- Perform an initial vacuum clearance for the entire system – Initial Vacuum Clearance
- Perform an individual vacuum clearance for sample pathways 1-3
    - Sample 1 gas cleanout
    - Sample 2 gas cleanout
    - Sample 3 gas cleanout
- Sample the gas to sample 1 P-10 tube – Sample 1 mix
- Sample the gas to sample 2 P-10 tube – Sample 2 mix
- Sample the gas to sample 3 P-10 tube – Sample 3 mix
- Sample the gas to MS tube – Sample to MS

HMI Usability

I designed the system's HMI to give operators clear options for running and understanding the automated sampling procedures. The main automation screen provides a guided mode and a manual mode, allowing users to choose the level of assistance they need. Guided mode launches a sequence of instructional popups that walks the users through the required actions while the PLC controls each automated process. Manual mode removes the step-by-step guidance and provides direct process-trigger buttons for operators already familiar with the system I also created separate control popups for manually operating individual valves and pumps during testing, troubleshooting, or maintenance.

Key Takeaways

This internship taught me that successful automation depends on more than writing control logic that works under the most ideal of conditions. I presented and demonstrated the outcomes of my project to USTC group members on the 25th of August 27, 2026. The most valuable lessons came from repeatedly testing the system, identifying unexpected timing behavior, and refining its sequences and operator controls. I also gained a strong understanding of how PLC scan cycle, HMI acquisition rates, actuators, sensors, and user inputs interact. Working independently while receiving guidance from my mentor and other more seasoned engineers improved my confidence, technical communication, and ability to solve unfamiliar engineering problems. Most importantly, the experience connected my computer science background with real

automation applications and confirmed my interest in pursuing controls engineering. This was a great experience all round and I feel like I was provided with so many avenues to grow as a professional.